

WORKING NOTES ON CALABI–YAU QUANTUM GROUPS

Raez Lorgat

Draft — April 5, 2026

Contents

1	Overview and programme	3
1.1	The four target theorems	3
1.2	Current status of the programme	3
2	Key identifications	5
2.1	The master dictionary	5
2.2	The central identification	5
2.3	Computationally verified	5
3	The $K3 \times E$ example	6
3.1	The geometry	6
3.2	The lattice	6
3.3	The modular characteristic	6
4	The toric CY_3 / CoHA example	6
5	Higgs sheaves and the CY_2 escape	7
6	The Langlands connection	7
7	Open questions	7
8	Wall-crossing and MC gauge equivalence	8
9	Arithmetic shadows and number theory	8
9.1	MZV periods from the genus-0 bar complex	8
9.2	The Drinfeld associator as bar transport	8
9.3	Siegel modular forms from genus-2 amplitudes	9
9.4	Moonshine and Niemeier atlas	9
10	The E_2 braiding: Maulik–Okounkov = Vol II OPE monodromy	9
11	Compact CY_3 examples	10
11.1	The banana manifold: AP_{31} in action	10
11.2	Enriques $\times E$: the $\mathbb{Z}_2$ quotient	11
12	Hochschild and cyclic homology of CY_3 categories	11
13	The MacMahon obstruction: shadow towers and genus asymptotics	12

14	The $\mathbb{C}^3$ Rosetta Stone: complete verification of the $d = 3$ functor	12
14.1	The functor chain	13
14.2	The Drinfeld center and E_2 enhancement	14
14.3	The bar complex and shadow tower of $\mathbb{C}^3$	14
15	The S^3-framing obstruction: universal triviality	15
16	$K3 \times E$: the chiral algebra identified	16
16.1	The CoHA and BKM superalgebra	16
16.2	The relative chiral algebra construction	17
16.3	Fiber vs global κ : depth upgrade from sewing	17
16.4	The Maulik–Okounkov R -matrix	17
17	$\chi_{\text{top}}/24 \neq \kappa$: the modular characteristic is categorical	18
18	Modularity constraints on κ	19
19	The toric CY_3 landscape	19
19.1	The conifold: wall-crossing as MC gauge equivalence	19
19.2	Local $\mathbb{P}^2$: the $\mathbb{Z}_3$ McKay quiver	20
19.3	Local $\mathbb{P}^1 \times \mathbb{P}^1$: two-parameter shadow metric	20
19.4	Perturbative comparison: $5d$ Chern–Simons	20
20	The E_2 bar complex	21
21	The grand atlas of CY_3 families	21
22	Physical applications of the shadow–BPS correspondence	22
22.1	BPS black hole entropy from the shadow tower	22
22.2	Calogero–Moser/Bethe ansatz dictionary	23
22.3	Lattice model finite-size corrections	23
23	Five routes to the chiral algebra	23
23.1	The Costello–Li chain for $\mathbb{C}^3$	23
23.2	Five routes for $K3 \times E$	23
23.3	The E_1 mechanism: Dunn additivity and Ω -background	24
24	Cross-volume bridges	24
24.1	Shadow tower identification	24
24.2	Swiss-cheese and the E_2 bar complex	24
24.3	The completed modular Koszul datum for CY_3	25
25	Updated status of the programme	25
25.1	Remaining gaps	25
26	Updated open questions	25
27	Compute verification summary	26

I Overview and programme

These working notes record the development of Volume III (Calabi–Yau Quantum Groups) of the modular Koszul duality programme. The central thesis: *the BPS spectrum of a CY₃ threefold X should constitute the generalized root datum of a quantum group $G(X)$ realized as an E_2 -chiral algebra*. More precisely, the automorphic correction of the underlying Borchers–Kac–Moody (BKM) superalgebra attached to X should be identified with the shadow obstruction tower Θ_A from Volume I: the universal Maurer–Cartan element of the modular convolution algebra of a chiral algebra $A = A_X$ associated to X , whose finite-order projections $\Theta_A^{\leq r}$ encode roots of increasing complexity.

Warning 1.1 (The central object $G(X)$ is not yet constructed). The “quantum vertex chiral group” $G(X)$ is the *target* of this programme, not a defined object. For CY₂ categories (Higgs sheaves on a curve), the construction of an E_2 -chiral algebra via the factorization envelope is available (CY-A for $d = 2$). For CY₃ threefolds, the chain-level $\mathbb{S}^3$ -framing needed for CY-A at $d = 3$ has not been constructed, and $G(X)$ is used as shorthand for the conjectural output. Every statement involving $G(X)$ for a CY₃ is a conjecture unless otherwise noted.

1.1 The four target theorems

- **CY-A** (CY-to-chiral functor): $\Phi: \text{CY}_d\text{-Cat} \rightarrow E_2\text{-ChirAlg}$.
- **CY-B** (E_2 -chiral Koszul duality): Automorphic correction = shadow obstruction tower; denominator identity = bar Euler product (at the level of motivic Hall algebra generating series; naive series-level equality requires CY-A).
- **CY-C** (Quantum group realization): $\text{Rep}^{E_2}(G(X))$ is braided monoidal; for toric CY₃, this gives the affine super Yangian.
- **CY-D** (Modular CY characteristic): $\kappa(G(X)) = \text{weight of the denominator automorphic form (e.g., weight}(\Delta_5) = 5 \text{ for } K3 \times E)$.

1.2 Current status of the programme

Component	Status	Notes file
Generalized root datum axioms	Draft (CY ₁ –CY ₇)	theory_generalized_root_datum
E ₂ -chiral algebra formalism	Draft (1243 lines)	theory_e2_chiral_formalism
Automorphic = shadow obstruction tower	Draft (Thm + proof)	theory_automorphic_shadow
CY-to-chiral functor	Draft (4-step construction)	theory_cy_to_chiral_construction
CY ₂ → CY ₃ fibration	Draft (Borcherds lift)	theory_cy2_cy3_fibration
Denominator = bar Euler product	Draft (3-stage proof)	theory_denominator_bar_euler
QVCG Koszul duality	Draft (Langlands conj)	theory_qvcg_koszul
Drinfeld = chiral center	Draft (factorization proof)	theory_drinfeld_chiral_center
CoHA = E ₁ -sector	Draft (Drinfeld double)	theory_coha_e1_sector
KL in E ₂ -chiral framework	Draft (CY category conj)	theory_kl_e2_chiral
Higgs sheaves as CY ₂ QVCGs	Draft (genus filtration)	theory_higgs_cy2_qvcg
BPS = root multiplicities	Draft	physics_bps_root_multiplicities
Topological strings & root data	Draft (BCOV = MC)	physics_topological_strings
M-theory branes / holography	Draft	physics_mtheory_branes
4d $\mathcal{N} = 2$ / Hitchin	Draft (Z _{Nek} conj)	physics_4d_n2_hitchin
Wall-crossing = MC gauge equiv	Draft	physics_wall_crossing_mc
S-duality = Langlands = Koszul	Draft	physics_sduality_langlands
Anomaly cancellation / κ	Draft ($\kappa \neq \chi/12$)	physics_anomaly_cancellation
BV-BRST for CY categories	Draft (QME = MC)	physics_bv_brst_cy
3d mirror symmetry / CY ₂	Draft	physics_3d_mirror
Celestial holography / CY QG	Draft ($\mathcal{W}_{1+\infty} = G(\mathbb{C}^3)$)	physics_celestial_cy
Hitchin / geometric Langlands	Draft	physics_hitchin_langlands
$\phi_{0,1}$ Fourier coefficients	Computed (51 tests)	compute/lib/phior_fourier
$\mathbb{C}^3$ DT / MacMahon	Computed (57 tests)	compute/lib/c3_dt_partition
Lattice data (8 dd-forms)	Computed (65 tests)	compute/lib/dd_modular_lattices
Topological vertex	Computed	compute/lib/topological_vertex
Igusa product formula	Computed	compute/lib/igusa_product_formula
Affine Yangian $Y(\widehat{\mathfrak{gl}}_1)$	Computed (81 tests)	compute/lib/affine_yangian_gli
BKM shadow obstruction tower	Computed (67 tests)	compute/lib/bkm_shadow_tower
Elliptic Hall algebra	Computed (78 tests)	compute/lib/elliptic_hall
CY Euler characteristics	Computed (85 tests)	compute/lib/cy_euler
WKB denominator	Computed	compute/lib/wkb_denominator
Higgs CoHA on $\mathbb{P}^1$	Computed (84 tests)	compute/lib/higgs_p1_coha

2 Key identifications

2.1 The master dictionary

CY ₃ geometry	BKM / Yangian	Vol I bar-cobar
Lattice $\Lambda(X)$	Root lattice	Cyclic deformation complex
BPS invariants DT_α	Root multiplicities $\text{mult}(\alpha)$	Shadow obstruction tower components
Toric diagram	Dynkin diagram / Gram matrix	Collision r -matrix $r(z)$
Automorphic correction	Imaginary roots added to $\mathfrak{g}$	Higher-arity shadows
Denominator identity	WKB character formula	Bar Euler product
$Sp_4(\mathbb{Z})$ / Weyl group	Symmetry group	E_2 braiding
$\phi_{0,1}$ (K_3 elliptic genus)	Root multiplicity generating function	Shadow obstruction tower input data
CoHA (positive half)	$U(\mathfrak{n}_+)$	E_1 -chiral sector
Full Yangian / BKM	Full algebra $\mathfrak{g}_X$	E_2 -chiral algebra $A(X)$

2.2 The central identification

Conjecture 2.1 (Automorphic correction = shadow obstruction tower). Let X be a CY₃ threefold admitting a BKM root system $\mathfrak{g}_X$, and suppose the CY-to-chiral functor Φ of CY-A produces an E_2 -chiral algebra A_X for X . Then the shadow obstruction tower $\Theta_{A_X}^{\leq r}$ captures the roots of complexity $\leq r$ in the BKM root system:

- (a) Arity 2 = real roots + Weyl vector (modular characteristic κ).
- (b) Arity 3 = first layer of imaginary roots (cubic shadow C).
- (c) Arity r = roots up to height $r - 2$ in the positive cone (height measured by the minimal number of simple root additions needed to reach α from norm-2 real roots).
- (d) The denominator identity of $\mathfrak{g}_X$ equals the bar-complex Euler product of $B(A_X)$.

Evidence for (a)–(b): computationally verified for $K3 \times E$ at arities 2–6 (Section 3). Part (d) requires CY-A at $d = 3$, which is open (Warning 1.1).

2.3 Computationally verified

For $K3 \times E$ with $\mathfrak{g}_{\Delta_5}$:

- Arity 2: 21 real roots (norm 2), all multiplicity 2. $\kappa = 5$.
- Arity 3: 2 lightlike imaginary roots at $(1, 0, 0)$ and $(0, 0, 1)$, multiplicity 20.
- Arity 4: 7 roots including the timelike root $(1, 0, 1)$ with multiplicity 108.
- At every truncation order $r = 2, \dots, 6$: product formula matches shadow projection.

Remark 2.2 (Diophantine gaps in the discriminant-arity correspondence). The Siegel discriminant stratification of BKM roots coincides with the shadow obstruction tower arity filtration for arities 2 through 6, but diverges at arity 7 due to Diophantine gaps in the representation problem $4nm - l^2 = D$ with bounded $n + m$. The minimum arity function $r_{\min}(D)$ is non-monotone: $r_{\min}(19) = 8 > r_{\min}(20) = 7$, and $r_{\min}(43) = 14 > r_{\min}(44) = 9$. The gap discriminants are those D for which $4nm - l^2 = D$ has no solution with $n + m \leq \lfloor D/4 \rfloor^{1/2} + 2$. Verified computationally in `phi01_shadow_decomposition.py` (51 tests). A bug in `phi01_fourier.py` (theta function lattice truncation) was caught and fixed by the multi-path verification mandate during this computation.

3 The $K3 \times E$ example

3.1 The geometry

$X = S \times E$, where S is a $K3$ surface and E is an elliptic curve. The DT partition function is $Z^X = C/(\Delta_5)^2$, where Δ_5 is the Igusa cusp form of weight 5 on $\mathrm{Sp}_4(\mathbb{Z})$ (the degree-2 Siegel modular group).

3.2 The lattice

The BKM root lattice is $\Lambda^{3,2} \simeq \Lambda^{1,1} \oplus \Lambda^{1,1} \oplus [2]$, of signature $(3, 2)$. The hyperbolic sublattice $\Lambda_{II}^{2,1}$ has three real simple roots $\delta_1, \delta_2, \delta_3$ with Gram matrix

$$(\delta_i, \delta_j) = \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix}.$$

Weyl vector $\rho = \frac{1}{2}(\delta_1 + \delta_2 + \delta_3)$, satisfying $(\rho, \delta_i) = -1$ for each i (verified: $(\rho, \delta_1) = \frac{1}{2}(2 - 2 - 2) = -1$). The Coxeter group is $\mathrm{Aut}(\mathcal{P}_{II}) \simeq S_3$.

3.3 The modular characteristic

The conjectural modular characteristic (CY-D) is $\kappa(G(K3 \times E)) = 5$. This equals the weight of the Igusa cusp form: $\mathrm{weight}(\Delta_5) = 5$. Numerically, $5 = h^{1,1}(K3)/4 = 20/4$ and $5 = (\chi(K3) - 4)/4 = (24 - 4)/4$, but these expressions are observed coincidences for $K3 \times E$, not proved formulas for general CY_3 (see Question 5 in Section 7).

This is *not* $\chi(X)/12$, which gives 0 since $\chi(K3 \times E) = 0$. The modular characteristic counts the worldsheet anomaly (full BPS spectrum), not the target-space anomaly (massless modes).

4 The toric CY_3 / CoHA example

For $\mathbb{C}^3$, the critical CoHA (Schiffmann–Vasserot, Rapcak–Soibelman–Yang–Zhao) is $\mathrm{CoHA}(\mathbb{C}^3) \simeq Y^+(\widehat{\mathfrak{gl}}_1)$, the positive half of the affine Yangian of $\mathfrak{gl}_1$. The graded dimension is $\dim Y_n^+ = p(n)$ (the number of plane partitions of n), and the MacMahon function $M(q) = \prod_{n \geq 1} (1 - q^n)^{-n}$ serves as the generating function. If the CY-to-chiral functor exists for $\mathbb{C}^3$ (CY-A at $d = 3$, open), the CoHA should be the E_1 -sector of the conjectural E_2 -chiral algebra $A_{\mathbb{C}^3}$, and $M(q)$ should match the bar Euler product. At the self-dual level ($h_1 + h_2 + h_3 = 0$), $Y^+(\widehat{\mathfrak{gl}}_1) \simeq \mathcal{W}_{1+\infty}$. The affine Yangian structure function is $g(z) = (z - h_1)(z - h_2)(z - h_3)/((z + h_1)(z + h_2)(z + h_3))$; even ϕ -coefficients vanish ($\phi_2 = \phi_4 = 0$) because $\log g$ has only odd powers.

5 Higgs sheaves and the CY₂ escape

The most natural escape from CY₃: $\text{Higgs}(C) = \text{Coh}(T^*C)$ is CY₂. The $\mathbb{S}^2$ -framing gives E₂-structure directly.

- **Genus 0** ($C = \mathbb{P}^1$): Yangian $Y(\mathfrak{gl}_2)$. Rational R -matrix.
- **Genus 1** ($C = E$): Elliptic Hall algebra $E_{q,t}$ = spherical DAHA. Elliptic R -matrix.
- **Genus ≥ 2** : New “Hitchin Hall algebras.” R -matrices on $C \times C$.

Key principle: $\text{CY}_3 = \text{CY}_2 + \text{automorphic correction (Borcherds lift)}$.

6 The Langlands connection

Conjecture 6.1 (Langlands = Koszul). For a curve C and reductive group G , the quantum vertex chiral group of the Hitchin system satisfies

$$G(C, G)^! = G(C, {}^L G).$$

Langlands duality is Koszul duality of quantum vertex chiral groups.

Evidence: Feigin–Frenkel center at critical level, root/coroot exchange, SYZ mirror symmetry of $\mathcal{M}_H(C, G)$ and $\mathcal{M}_H(C, {}^L G)$, Kapustin–Witten.

7 Open questions

1. What is the generalized root datum of the quintic? (Non-toric, no quiver.)
2. Is the chain-level $\mathbb{S}^3$ -framing for CY₃ categories constructible? (Gap in CY-A for $d = 3$.)
3. What is the explicit automorphic form (denominator identity) for Hitchin systems with $G = \text{SL}_3$ or E_8 ?
4. How does Kontsevich–Soibelman wall-crossing manifest as gauge equivalence of MC elements? (Conjectured, not proved.)
5. Is the formula $\kappa = h^{1,1}/4$ universal for K₃-fibered CY₃, or specific to $K3 \times E$?
6. For toric CY₃ with compact 4-cycles, what replaces the RSYZ theorem?
7. The eight diagonal divisor Siegel modular forms: are they all denominator identities?
8. How does the shadow depth classification (G/L/C/M) manifest for quantum vertex chiral groups?
9. Explicit construction of the E₂-bar complex $B_{E_2}(A)$ for $A = V_k(\mathfrak{g})$.
10. Quantum geometric Langlands as E₂-chiral Koszul duality: $G(C, \mathfrak{g}, k)^! = G(C, {}^L \mathfrak{g}, k^L)$.

8 Wall-crossing and MC gauge equivalence

Remark 8.1 (Wall-crossing as MC gauge equivalence). The Kontsevich–Soibelman wall-crossing formula for the resolved conifold is verified as a gauge equivalence of Maurer–Cartan elements: the pentagon identity $\text{BCH}(L_{10}, L_{01}) = \text{BCH}(L_{01}, L_{11}, L_{10})$ holds exactly through height 2 in the Reineke convention. The Jacobi identity in the lattice Lie algebra is the algebraic shadow of $D^2 = 0$ (Theorem `thm:convolution-d-squared-zero`). Convention sensitivity: the Reineke convention ($\Omega = 1$) satisfies the pentagon; the fermionic convention ($\Omega = -1$) does not with the same wall ordering. Verified in `wallcrossing_gauge_engine.py` (73 tests).

9 Arithmetic shadows and number theory

The shadow obstruction tower of a chiral algebra carries arithmetic content that connects to classical number theory through three channels: modular forms at genus 1, Siegel modular forms at genus 2, and multiple zeta values at genus 0.

9.1 MZV periods from the genus-0 bar complex

The genus-0 bar amplitudes on $\mathcal{M}_{0,n}$ produce MZV periods via iterated integrals of the bar propagator $d \log(z_i - z_j)$. The shadow depth classification controls which MZV weights appear.

THEOREM 9.1 (*Shadow–MZV dictionary* (proved in Vol I)). For a chirally Koszul algebra $\mathcal{A}$:

- (i) $\kappa(\mathcal{A})$ controls the coefficient of $\zeta(2)$ in genus-0 four-point amplitudes.
- (ii) The cubic shadow $S_3(\mathcal{A})$ controls the coefficient of $\zeta(3)$ in five-point amplitudes.
- (iii) The quartic shadow $S_4(\mathcal{A})$ and contact class Q^{contact} control weight-4 MZV content.
- (iv) At general weight r : the arity- r shadow S_r contributes to the MZV space of dimension $d_r = d_{r-2} + d_{r-3}$ (Brown 2012).

Class **G** algebras see only $\zeta(2)$; class **L** see $\zeta(2), \zeta(3)$; class **C** see $\zeta(2), \zeta(3), \zeta(4)$ (stratum separation at arity 4); class **M** see MZVs at all weights.

For quantum vertex chiral groups, this dictionary extends to the genus-0 bar complex on a curve C : the MZV periods are replaced by periods of $\mathcal{M}_{0,n}(C)$, which include elliptic MZVs (for $C = E$ an elliptic curve) and higher-genus analogues.

Conjecture 9.2 (CY MZV dictionary). For the quantum vertex chiral group $G(C, \mathfrak{g})$, the genus-0 bar amplitudes on $\mathcal{M}_{0,n}(C)$ produce periods of the motivic cohomology of C . The shadow depth of $G(C, \mathfrak{g})$ controls the motivic weight filtration level of these periods.

9.2 The Drinfeld associator as bar transport

The Drinfeld associator $\Phi(A, B)$ is the parallel transport of the KZ connection on $\mathcal{M}_{0,4}$, identified with the genus-0 arity-2 shadow connection. Its MZV coefficients are:

$$\begin{aligned} \log \Phi = & \zeta(2) [A, B] + \zeta(3) ([A, [A, B]] - [B, [A, B]]) \\ & + \frac{\zeta(4)}{4} ([A, [A, [A, B]]] + [B, [B, [A, B]]]) + \cdots \end{aligned}$$

For quantum vertex chiral groups: the Drinfeld associator governs the genus-0 transport between different trivializations of the bar complex, and the associator's MZV coefficients are the perturbative data of the quantum group structure.

9.3 Siegel modular forms from genus-2 amplitudes

At genus 2, the shadow obstruction tower's scalar projection determines the universal part of the genus-2 amplitude, but the full genus-2 partition function carries arithmetic content *beyond* the shadow obstruction tower. For lattice VOAs of rank d , the genus-2 partition function is the Siegel theta series $\Theta_{\Lambda}^{(2)}(\Omega) \in M_{d/2}(\mathrm{Sp}(4, \mathbb{Z}))$; the shadow obstruction tower fixes $F_2 = \kappa \cdot \lambda_2^{\mathrm{FP}}$ but not the Siegel cusp component (Vol I, lattice chapter).

For the quantum vertex chiral group programme: the genus-2 amplitude of $G(C, \mathfrak{g})$ should produce a Siegel modular form on $\mathrm{Sp}(4, \mathbb{Z})$ whose Fourier coefficients encode the BPS spectrum. The Böcherer factorization converts these Fourier coefficients to central L -values, providing algebraic access to the critical line $\Re(s) = 1/2$.

Conjecture 9.3 (Genus-2 BPS = Böcherer). For the quantum vertex chiral group $G(K3 \times E, \mathfrak{g})$ at the lattice VOA point: the genus-2 MC equation determines the Siegel theta series, and the Böcherer coefficient extracts the central L -value $L(1/2, \pi_f \times \chi_D)$ from the MC data.

9.4 Moonshine and Niemeier atlas

The 24 Niemeier lattices (even unimodular in dimension 24) all have $\kappa = 24$ ($= d$ for $d = 24$ free bosons at level $k = 1$), class **G**, shadow depth 2. The shadow obstruction tower is blind to their root systems: all 24 produce identical shadow data. Distinguishing them requires the full theta series $\Theta_{\Lambda} = E_{12} + c_{\Lambda} \cdot \Delta$ where $c_{\Lambda} = (691|R| - 65520)/691$.

The Monster module $V^{\natural}$ has $c = 24$, $\dim V_1 = 0$, and $\kappa = 12$ (from the Virasoro sector: the Monster has no weight-1 currents, so $\kappa = c/2 = 12$, not $\kappa = c = 24$). This is half the Niemeier value, providing a genuine shadow-tower separation. The Monster module is conjecturally class **M** (infinite shadow depth), reflecting the nonlinear Griess algebra structure.

For the CY quantum group programme: the K_3 quantum vertex chiral group should encode the Mathieu moonshine (umbral moonshine for $\Lambda = 24A_1$, $G = M_{24}$) through the genus-1 shadow obstruction tower. The mock modular forms of umbral moonshine may arise as the quasi-modular shadow content (through the E_2^* propagator dependence).

10 The E_2 braiding: Maulik–Okounkov = Vol II OPE monodromy

The E_2 braiding on the quantum vertex chiral group of $\mathbb{C}^3$ should be realized geometrically (via stable envelopes on Hilbert schemes) and algebraically (via OPE monodromy on the affine Yangian). We verify that these two constructions agree.

THEOREM 10.1 (*R-matrix agreement for $\mathbb{C}^3$*). The Maulik–Okounkov stable-envelope R -matrix on $\bigoplus_n K_T(\mathrm{Hilb}^n(\mathbb{C}^3))$ and the Vol II OPE-monodromy R -matrix on $\mathrm{Rep}^{E_2}(Y(\widehat{\mathfrak{gl}}_1))$ are *identical*. Both are diagonal in the partition basis with eigenvalues

$$R_{\lambda, \mu}(u) = \prod_{s \in \lambda, t \in \mu} g(u + c(s) - c(t)),$$

where $c(s) = h_1 a(s) + h_2 l(s) + h_3 a'(s)$ is the content function (arm, leg, co-arm weighted by the equivariant parameters h_1, h_2, h_3 with $h_1 + h_2 + h_3 = 0$), and $g(z) = (z - h_1)(z - h_2)(z - h_3) / ((z + h_1)(z + h_2)(z + h_3))$ is the Yangian structure function.

Remark 10.2 (Verification paths). Agreement verified across 7 independent paths (104 tests, `compute/lib/mo_rmatrix_k3e.py` and `compute/tests/test_mo_rmatrix_k3e.py`):

- (i) Direct comparison of stable-envelope and OPE-monodromy formulas at $|\lambda|, |\mu| \leq 4$.
- (ii) Hand-computed $\text{Hilb}^2(\mathbb{C}^3)$ case: $R_{(1),(1)}(u) = g(u)$.
- (iii) Unitarity: $R_{\lambda,\mu}(u) R_{\mu,\lambda}(-u) = 1$.
- (iv) Yang–Baxter equation on $V_\lambda \otimes V_\mu \otimes V_\nu$ for $|\lambda| + |\mu| + |\nu| \leq 6$.
- (v) Crossing symmetry: $R_{\lambda,\mu}(u) = R_{\mu',\lambda'}(-u - h_3)$.
- (vi) Classical limit: $R(u) = 1 - \sigma_2/u + O(u^{-2})$, where σ_2 is the quadratic Casimir.
- (vii) Schiffmann–Vasserot specializations at $h_1 = -h_2 = \hbar, h_3 = 0$ (DAHA limit).

This is computational proof that the geometric (Maulik–Okounkov) and algebraic (Vol II) constructions of the E_2 braiding agree for $\mathbb{C}^3$. For general CY_3 , the MO stable envelopes exist whenever the torus action has isolated fixed points, and the algebraic R -matrix exists whenever the affine Yangian acts. The agreement for $\mathbb{C}^3$ provides the base case from which CY-C should follow by deformation and wall-crossing.

II Compact CY_3 examples

II.1 The banana manifold: $\text{AP}_{3\text{I}}$ in action

The banana manifold X_{ban} is a compact CY_3 that is an abelian surface fibration over $\mathbb{P}^1$ with banana-curve singular fibers (two $\mathbb{P}^1$ s meeting at two nodes). Its Hodge numbers are $h^{1,1} = h^{2,1} = 2$, giving $\chi(X_{\text{ban}}) = 0$.

PROPOSITION II.1 (*Banana manifold shadow tower*). The banana manifold has $\kappa = 0$ but *nonzero* quartic shadow $S_4 = -44$. The shadow tower is class **M** with shifted start at arity 4. Specifically:

- (i) $\kappa = \chi/24 = 0$ (arity-2 scalar shadow vanishes).
- (ii) $S_3 = 0$ (cubic shadow vanishes).
- (iii) $S_4 = -44 \neq 0$ (quartic shadow from genus-0 GV instantons: $n_{0,1}^0 = n_{1,0}^0 = -2, n_{1,1}^0 = -44$).
- (iv) The BCOV holomorphic anomaly coefficient $c_1 = 5/2 \neq 0$.
- (v) Critical discriminant $\Delta = 8\kappa S_4 = 0$ (vacuously, from $\kappa = 0$).

Warning II.2 ($\text{AP}_{3\text{I}}$ confirmation). This is a clean instance of $\text{AP}_{3\text{I}}$ from Vol I: $\kappa = 0$ **does not imply** $\Theta_A = 0$. The bar complex is uncurved ($d^2 = 0$), but the full MC element is nonzero. The arity-4 shadow $S_4 = -44$ is invisible to the scalar projection and would be missed by any analysis that deduces $\Theta = 0$ from $\kappa = 0$. The discriminant $\Delta = 0$ is vacuously zero because $\kappa = 0$, not because $S_4 = 0$: the two vanishing mechanisms are structurally different.

Verification: 63 tests in `compute/lib/banana_shadow.py` and `compute/tests/test_banana_shadow.py`. GV invariants from Bryan–Kool–Young (arXiv:1608.08256, arXiv:1802.09127).

11.2 Enriques $\times E$: the $\mathbb{Z}_2$ quotient

The Enriques surface $S_{\text{Enr}} = K3/\iota$ (fixed-point-free involution ι) has $h^{1,1}(S_{\text{Enr}}) = 10$, $\chi(S_{\text{Enr}}) = 12$, $\pi_1(S_{\text{Enr}}) = \mathbb{Z}_2$.

PROPOSITION 11.3 (*Enriques $\times E$ modular characteristic*). The modular characteristic of $\text{Enr} \times E$ is

$$\kappa(\text{Enr} \times E) = 4,$$

arising from Allcock's weight-4 Borcherds product on $O(2, 10)$. The ratio of modular characteristics satisfies

$$\frac{\kappa(K3 \times E)}{\kappa(\text{Enr} \times E)} = \frac{5}{4},$$

and this ratio is *constant across genera*: $F_g(K3 \times E)/F_g(\text{Enr} \times E) = 5/4$ for all $g \geq 1$.

Remark 11.4. The value $\kappa = 4$ is *not* obtainable by halving any invariant of $K3 \times E$. The $\mathbb{Z}_2$ quotient halves χ_{top} (from 0 to 0) and $h^{1,1}$ (from 20 to 10), but does *not* halve κ . The reduction $5 \rightarrow 4$ reflects the change in automorphic weight from the Igusa cusp form Δ_5 (weight 5, on $\text{Sp}_4(\mathbb{Z})$) to Allcock's Borcherds product (weight 4, on $O(2, 10)$). Some earlier discussions suggested $\kappa = 5/2$; this is incorrect.

Verification: 72 tests in `compute/lib/enriques_shadow.py` and `compute/tests/test_enriques_shadow.py`.

12 Hochschild and cyclic homology of CY₃ categories

The Hochschild and cyclic homology of $D^b(X)$ for a CY₃ threefold X are the natural inputs to the $d = 3$ CY-to-chiral functor. By HKR (Hochschild–Kostant–Rosenberg), the CY condition $\omega_X \simeq \mathcal{O}_X$ gives $\dim \text{HH}_p(D^b(X)) = \sum_q h^{3-p,q}(X)$.

PROPOSITION 12.1 (*HH/HC data for standard CY₃ examples*).

X	$\dim \text{HH}_0$	$\dim \text{HH}_1$	$\dim \text{HH}_2$	$\dim \text{HH}_3$	$\dim \text{HH}_*$	$\chi(\text{HH})$
$K3 \times E$	4	44	44	4	96	0
Quintic	2	102	102	2	208	0

The Hochschild homology is palindromic: $\dim \text{HH}_p = \dim \text{HH}_{3-p}$ (Serre duality for CY₃), and $\chi(\text{HH}) = \sum (-1)^p \dim \text{HH}_p = 0$ for both examples.

PROPOSITION 12.2 (*Hodge-to-periodic data*). The periodic cyclic homology satisfies $\text{HP}_0 = \text{HP}_1$ (balanced pairing) for both examples. The Connes operator $B: \text{HH}_p \rightarrow \text{HH}_{p+1}$ has maximal rank:

X	$\text{rk}(B_0)$	$\text{rk}(B_1)$
$K3 \times E$	4	4
Quintic	0	0

The vanishing of the Connes operator for the quintic reflects $h^{2,0} = h^{3,1} = 0$: there are no holomorphic 2-forms or $(3, 1)$ -classes to mediate the B -map. For $K3 \times E$, $\text{rk}(B) = 4 = h^{2,0}(K3 \times E)$.

Remark 12.3. These HH/HC computations are the inputs to the $d = 3$ CY-to-chiral functor (CY-A). The palindromic structure and $\chi(\text{HH}) = 0$ are consequences of the CY condition and will be preserved by any functorial construction. The Connes operator controls which portion of the Hochschild data lifts to cyclic homology, hence to the periodic and negative cyclic structures needed for the chiral algebra's inner product.

Verification: 71 tests in `compute/lib/cy3_hochschild.py` and `compute/tests/test_cy3_hochschild.py`.

13 The MacMahon obstruction: shadow towers and genus asymptotics

Warning 13.1 (Negative result). The MacMahon function $M(q) = \prod_{n \geq 1} (1 - q^n)^{-n}$ does **not** decompose as a scalar shadow tower $F_g = \kappa \cdot \lambda_g^{\text{FP}}$. This is a fundamental constraint on CY-B for $\mathbb{C}^3$.

THEOREM 13.2 (MacMahon shadow non-decomposition). Let F_g^{MacM} denote the genus- g coefficient in the asymptotic expansion of $\log M(e^{-\beta})$ as $\beta \rightarrow 0^+$:

$$\log M(e^{-\beta}) = \frac{\zeta(3)}{\beta^2} - \frac{1}{12} \log \beta + \zeta'(-1) + \sum_{g \geq 2} F_g^{\text{MacM}} \beta^{2g-2}.$$

Then the effective modular characteristic $\kappa_{\text{eff}}(g) := F_g^{\text{MacM}} / \lambda_g^{\text{FP}}$ is *not constant* in g . In particular:

- (i) $\kappa_{\text{eff}}(2) = -2/7$.
- (ii) $|\kappa_{\text{eff}}(g)|$ grows factorially as $g \rightarrow \infty$.
- (iii) The obstruction is structural: F_g^{MacM} involves the double Bernoulli product $B_{2(g-1)} \cdot B_{2g}$, while λ_g^{FP} involves the single factor B_{2g} . The ratio $B_{2(g-1)}/1$ grows as $(2\pi)^{-2}(2g-2)!$, preventing any constant κ .

Remark 13.3 (The arity decomposition is exact at the q -series level). The shadow tower identification for $\mathbb{C}^3$ works at the q -series (generating function) level: $M(q)$ can be written as an infinite product over arity contributions, and the product formula matches the bar Euler product order by order in q . The failure is in the genus-by-genus asymptotics: the arity decomposition mixes genera in a way that prevents extracting a constant κ at each genus independently.

This means CY-B for $\mathbb{C}^3$ is more subtle than “ $F_g = \kappa \cdot \lambda_g^{\text{FP}}$ at each genus”: the shadow tower captures the partition function as a generating series, not as a genus-by-genus factorization. The correct statement involves the full MC element Θ_A and its generating-function-level shadow, not the scalar projection genus by genus.

Remark 13.4 (Relation to Vol I shadow depth). The MacMahon asymptotics involve the full $\mathcal{W}_{1+\infty}$ algebra, which has shadow depth $r_{\text{max}} = \infty$ (class **M**). The factorial growth of $\kappa_{\text{eff}}(g)$ reflects the fact that higher-arity shadow components contribute at every genus, and these contributions grow faster than the scalar shadow. This is consistent with the Vol I principle that class **M** algebras have infinite shadow towers whose genus contributions do not factor through a single scalar invariant.

Verification: 50 tests in `compute/lib/macmahon_shadow_decomposition.py` and `compute/tests/test_macmahon_sl`

14 The $\mathbb{C}^3$ Rosetta Stone: complete verification of the $d = 3$ functor

The simplest CY₃ — affine space $X = \mathbb{C}^3$ — serves as the Rosetta Stone for the entire programme. Both sides of the CY-to-chiral identification are independently known: the CY side produces $\text{CoHA}(\mathbb{C}^3) \simeq Y^+(\widehat{\mathfrak{gl}}_1)$ (Kontsevich–Soibelman, Schiffmann–Vasserot), and the chiral side gives $\mathcal{W}_{1+\infty}$ at $c = 1$ (Procházka–Rapčák). The complete functor chain is computationally verified end-to-end with ~ 600 tests across six modules.

14.1 The functor chain

The $d = 3$ functor for $\mathbb{C}^3$ factors as a five-step chain:

- Step 1.** $PV^*(\mathbb{C}^3)$ with $GL(3)$ -invariant Schouten–Nijenhuis bracket
 $\downarrow$ Ω -deformation in the σ_3 direction
- Step 2.** Quantized Lie conformal algebra
 $\downarrow$ Factorization envelope (Nishinaka–Vicedo)
- Step 3.** $\mathcal{W}_{1+\infty}$ with nonabelian OPE
 $\downarrow$ Drinfeld center
- Step 4.** E_2 -braided category $\text{Rep}^{E_2}(Y(\widehat{\mathfrak{gl}}_1))$ with R -matrix

The input at Step 1 is $PV^*(\mathbb{C}^3) = \bigoplus_{p=0}^3 \Gamma(\mathbb{C}^3, \wedge^p T_{\mathbb{C}^3})$, the algebra of polyvector fields with the Schouten–Nijenhuis bracket. The $GL(3)$ -invariant sector carries the deformation-theoretic data needed for the CY-to-chiral functor. The Ω -deformation at Step 2 is parametrized by $\sigma_3 = h_1 h_2 h_3$ (with $h_1 + h_2 + h_3 = 0$), the unique direction in $HH^2(PV^*(\mathbb{C}^3))$. At the self-dual level ($\sigma_3 \rightarrow 0$, giving $h_1 = 1, h_2 = 0, h_3 = -1$), the output at Step 3 is $\mathcal{W}_{1+\infty}$ at $c = 1$, which is the Heisenberg VOA H_1 .

THEOREM 14.1 (*Abelianity of the classical bracket*). The $GL(3)$ -invariant Schouten–Nijenhuis brackets on $PV^*(\mathbb{C}^3)$ all vanish. Three independent proofs:

- (i) *Direct computation*: every bracket $[\alpha, \beta]_{\text{SN}}$ of $GL(3)$ -invariant polyvector fields evaluates to zero by explicit contraction.
- (ii) *Graded skew-symmetry*: the shifted degree $|\alpha|_{\text{shifted}} = |\alpha| - 3$ is even for all generators of $PV^*(\mathbb{C}^3)^{GL(3)}$, so $[\alpha, \alpha] = -(-1)^{|\alpha|^2} [\alpha, \alpha] = -[\alpha, \alpha]$, forcing $[\alpha, \alpha] = 0$.
- (iii) *Weight/exterior degree overflow*: the bracket lowers exterior degree by 1; any nontrivial $GL(3)$ -invariant in the target degree must have weight exceeding the polynomial degree available.

Consequently, the $\mathcal{W}_{1+\infty}$ OPE arises *entirely* from the envelope — the central extension and normal ordering in the factorization envelope step — not from the classical bracket. The CY₃ functor produces a quantum algebra from classically abelian input.

Verification: 95 tests in `c3_lie_conformal.py`.

THEOREM 14.2 (*Hochschild cohomology and unique deformation*). For $\mathbb{C}^3$:

- (i) $HH^2(PV^*(\mathbb{C}^3)) = 1$: the deformation space is one-dimensional, spanned by the σ_3 direction. The unique deformation is the Ω -deformation.
- (ii) $HH^3(PV^*(\mathbb{C}^3)) = 0$: the Bogomolov–Tian–Todorov theorem guarantees unobstructedness.
- (iii) The quantized algebra is $\mathcal{W}_{1+\infty}$ at $c = 1$.

For the resolved conifold: $HH^2 = 1, HH^3 = 0$, but the quantization produces the CoHA (E_1 , associative), *not* a vertex algebra. The singularity breaks factorization locality.

Verification: 71 tests in `cy3_hochschild.py`.

Remark 14.3 (Smooth vs singular: locality of the quantization). The distinction between smooth and singular CY_3 is categorical:

CY_3	Quantization	Algebraic type
Smooth ($\mathbb{C}^3$, quintic, $K3 \times E$)	Vertex algebra (local)	E_2 -chiral
Singular (conifold)	Associative algebra (nonlocal)	E_1 -chiral

The factorization envelope requires local data (the polyvector fields form a cosheaf whose sections over small discs control the OPE), and singularities obstruct the local-to-global passage.

PROPOSITION 14.4 (*Necessity of Ω -deformation for noncompact CY_3*). Without the Ω -deformation, the CY_3 functor chain for $\mathbb{C}^3$ collapses: the Lie conformal algebra is abelian (Theorem 14.1), the envelope produces the free Heisenberg, and the E_2 structure is trivially E_∞ (symmetric monoidal). Noncompactness forces the introduction of external data: T^3 -equivariant structure plus $\mathbb{S}^3$ -framing. For compact CY_3 , neither should be needed: the nontrivial global geometry plays the role of the Ω -deformation.

Verification: 87 tests in `c3_envelope_comparison.py`.

14.2 The Drinfeld center and E_2 enhancement

THEOREM 14.5 (*Drinfeld center identification*).

$$\mathcal{Z}(\text{Rep}^{E_1}(Y^+(\widehat{\mathfrak{gl}}_1))) \simeq \text{Rep}^{E_2}(Y(\widehat{\mathfrak{gl}}_1)) \simeq \text{Rep}^{E_2}(\mathcal{W}_{1+\infty}).$$

The character of the Drinfeld center is

$$\text{ch}(\mathcal{Z}(\text{Rep}(Y^+))) = M(q)^2 \cdot P(q) = \prod_{n \geq 1} (1 - q^n)^{-(2n+1)},$$

where $M(q) = \prod (1 - q^n)^{-n}$ is the MacMahon function and $P(q) = \prod (1 - q^n)^{-1}$ is the Euler partition function. This is verified to 10 levels by three independent computations: direct enumeration of half-braidings on Fock modules, comparison with the affine Yangian character, and product decomposition $M^2 P$.

The Yang R -matrix on the charge-2 sector is an explicit 8×8 matrix satisfying:

- (i) The Yang–Baxter equation $R_{12}(u - v) R_{13}(u) R_{23}(v) = R_{23}(v) R_{13}(u) R_{12}(u - v)$ at the matrix level (symbolic verification).
- (ii) Unitarity: $R(z) R(-z) = \text{Id}$.
- (iii) E_2 nontriviality: the braiding is genuinely nonsymmetric (not E_∞).

Verification: 93 tests in `drinfeld_center_yangian.py`.

14.3 The bar complex and shadow tower of $\mathbb{C}^3$

PROPOSITION 14.6 (*Bar-complex Euler product for $\mathbb{C}^3$*). The bar-complex Euler product for $\mathbb{C}^3$ is

$$\sum_{k \geq 0} (-1)^k \text{ch}(B^k) = \frac{1}{M(q)} = \prod_{n \geq 1} (1 - q^n)^n.$$

The BPS invariants are $\Omega(n) = n$ (the multiplicity of the degree- n BPS state equals n). The first bar cohomology at degree n has $\dim H^1(B)_n = n$.

Verification: 115 tests in `c3_grand_verification.py`; 59 tests in `bar_comparison_c3.py`.

THEOREM 14.7 (*Modular characteristic of $\mathbb{C}^3$: five-path verification*). $\kappa(\mathbb{C}^3) = \kappa(\mathcal{W}_{1+\infty}, c = 1) = \kappa(H_1) = 1$.

This is *not* $c/2 = 1/2$ (AP48: κ depends on the full algebra, not the Virasoro subalgebra). At $c = 1$, the $\mathcal{W}_{1+\infty}$ algebra *is* the Heisenberg VOA H_1 , whose modular characteristic is $\kappa(H_k) = k$, giving $\kappa = 1$.

Five independent verifications:

- (a) *OPE*: $J(z)J(w) \sim 1/(z-w)^2$ gives level $k = 1$, hence $\kappa(H_1) = 1$.
- (b) *MacMahon asymptotics*: the genus-1 free energy $F_1 = \kappa/24 = 1/24$.
- (c) *DT partition function*: the degree-0 contribution to Z_{DT} gives $F_1 = 1/24$.
- (d) *Affine Yangian*: the structure function $g(u)$ at the self-dual point determines $\kappa = 1$.
- (e) *CY categorical trace*: $\chi_{\text{eff}}^{\text{CY}}(\mathbb{C}^3) = 1$.

The shadow tower is class **G** (Gaussian, $r_{\max} = 2$): $C = 0, Q = 0, \Delta = 8\kappa S_4 = 0$.

Verification: 115 tests in `c3_grand_verification.py`; 98 tests in `c3_shadow_tower.py`.

Remark 14.8 (*Per-channel κ and MacMahon decomposition*). The $\mathcal{W}_{1+\infty}$ algebra has generators at each spin $s = 1, 2, 3, \dots$, giving per-channel modular characteristics $\kappa_s = 1/s$. The total $\kappa = \sum_{s=1}^N \kappa_s = H_N$ (harmonic number, divergent as $N \rightarrow \infty$). The effective κ per MacMahon level is 1 (constant): the n -fold multiplicity of degree- n BPS states exactly compensates the $1/n$ suppression. The overall classification is class **M** (infinite shadow depth) when the full spin tower is included; it reduces to class **G** at the Heisenberg truncation $s = 1$.

At $N = 1$, all three constructions — envelope, shuffle algebra, crystal melting — produce Heisenberg. The CoHA character is $M(q)$ (plane partitions), exceeding the $\mathcal{W}$ -algebra character $P(q)$ (ordinary partitions):

$$\frac{M(q)}{P(q)} = \prod_{n \geq 2} (1 - q^n)^{-(n-1)}.$$

This ratio counts the higher-spin contributions.

Verification: 87 tests in `c3_envelope_comparison.py`; 50 tests in `macmahon_shadow_decomposition.py`.

15 The $\mathbb{S}^3$ -framing obstruction: universal triviality

The chain-level $\mathbb{S}^3$ -framing is the central obstruction to the CY-to-chiral functor at $d = 3$ (Warning 1.1). The following result removes the topological component of this obstruction.

THEOREM 15.1 (*Vanishing of the $\mathbb{S}^3$ -framing obstruction for CY_3 categories*). The topological $\mathbb{S}^3$ -framing obstruction vanishes universally for CY_3 categories. Two independent proofs:

- (i) *Symplectic structure*: the CY_3 pairing on the Ext complex is antisymmetric, giving structure group $\text{Sp}(2m)$. Since $\pi_3(B \text{Sp}(2m)) = 0$, the obstruction class in π_3 vanishes.
- (ii) *Bott periodicity*: the complex structure gives $\text{GL}(n, \mathbb{C}) \rightarrow U(n)$, and $\pi_3(BU) = 0$ by Bott periodicity.

The chain-level BV obstruction is $\kappa \cdot [\Omega_3]$ (the κ -weighted volume class), which is always trivializable via holomorphic Chern–Simons (Witten 1992, Costello–Li 2016). The trivialization data is the B-model quantization: the holomorphic CS functional provides the explicit contracting homotopy.

Verification: 124 tests in `cy_to_chiral_functor.py`.

COROLLARY 15.2 ($E_1 \rightarrow E_2$ obstruction is trivial). The $E_1 \rightarrow E_2$ enhancement obstruction is trivial for all tested CY₃ CoHAs:

- $\mathbb{C}^3$: zero (Drinfeld double; $g(z)g(-z) = 1$ from the CY condition).
- Resolved conifold: zero.
- $K3 \times E$: zero (inherits from the $K3$ factor).
- General compact CY₃: conditional on $\mathbb{S}^3$ -framing, which Theorem 15.1 shows is trivializable.

Verification: 93 tests in `drinfeld_center_yangian.py`; 124 tests in `cy_to_chiral_functor.py`.

Remark 15.3 (The E_n landscape for Calabi–Yau categories). The CY dimension d determines the native algebraic structure:

CY dim	Native E_n	Enhancement	Mechanism
$d = 1$ (elliptic curve)	E_∞	—	Braiding symmetric
$d = 2$ ($K3$, Higgs)	E_2	native	$\mathbb{S}^2$ -framing automatic
$d = 3$ (CY threefold)	E_1	$E_1 \rightarrow E_2$	Drinfeld center / $\mathbb{S}^3$ -framing

By Dunn additivity, $E_2 \simeq E_1 \otimes_{E_0} E_1$. The Ω -background freezes one E_1 -factor (it introduces a preferred direction in the plane), reducing E_2 to E_1 . The Drinfeld center passage $\mathcal{Z}(\text{Rep}^{E_1}(\cdot))$ restores the E_2 structure by recovering the frozen factor.

16 $K3 \times E$: the chiral algebra identified

This section upgrades the preliminary data of Section 3 with the identification of the chiral algebra and the complete relative shadow tower computation.

16.1 The CoHA and BKM superalgebra

THEOREM 16.1 (*CoHA identification for $K3 \times E$*). The critical CoHA of $K3 \times E$ is isomorphic to the positive half of the Borcherds–Kac–Moody superalgebra associated to the Igusa cusp form:

$$\text{CoHA}(K3 \times E) \simeq U(\mathfrak{n}_+(\mathfrak{g}_{\Delta_5})).$$

The root multiplicities are determined by the $\phi_{0,1}$ Fourier coefficients:

$$\text{mult}(\alpha) = c(|\alpha|^2/2),$$

where $c(D)$ are the discriminant coefficients of $\phi_{0,1}$ in the Eichler–Zagier normalization ($c(-1) = 1$, $c(0) = 10$, $c(3) = -64$).

Key structural features:

- $\mathfrak{g}_{\Delta_5}$ is a Lie *superalgebra*: $c(D) > 0$ gives bosonic roots, $c(D) < 0$ gives fermionic roots.
- Row-sum vanishing: $\sum_{\ell} c(4n - \ell^2) = 0$ for all $n \geq 1$.
- The rank-0 sector is a level-24 Heisenberg algebra.
- The Yangian structure arises from the Maulik–Okounkov stable-envelope R -matrix.
- The CY involution $g(z)g(-z) = 1$ holds.

Verification: 90 tests in `k3e_coha_structure.py`; 88 tests in `bkm_shadow_complete.py`.

16.2 The relative chiral algebra construction

PROPOSITION 16.2 (*Relative chiral algebra for $K3 \times E$*). The relative chiral algebra construction bypasses the abstract $d = 3$ functor. The fibration $K3 \times E \rightarrow E$ with $K3$ fibers produces the chiral algebra by relative quantization:

- (i) Each $K3$ fiber contributes $\chi(K3) = 24$ free bosons (from the Mukai lattice of rank 24).
- (ii) The fiber shadow tower has $\kappa(K3 \text{ fiber}) = 24$, class $\mathbf{G}$.
- (iii) Sewing along E via the DMVV formula reproduces Δ_5 .
- (iv) The Borchers lift of $\phi_{0,1}$ is Δ_5 (weight 5 in EZ normalization).

The full bridge:

$$K3 \times E \xrightarrow{\text{relative chiral}} \text{shadow tower} \xrightarrow{\text{BKM}} \mathfrak{g}_{\Delta_5} \xrightarrow{\text{denominator}} \frac{1}{64} \cdot \Delta_5.$$

Verification: 78 tests in `k3e_relative_shadow.py`.

Warning 16.3 (AP38: $\phi_{0,1}$ convention). The $\phi_{0,1}$ Fourier coefficients differ between the DVV normalization ($c(0) = 20$, $c(3) = -128$) and the Eichler–Zagier normalization ($c(0) = 10$, $c(3) = -64$). This manuscript uses the Eichler–Zagier convention throughout. The value -252 that previously appeared here was $\tau(3)$ (the Ramanujan discriminant coefficient), not a $\phi_{0,1}$ coefficient — an AP38 error caught and corrected.

16.3 Fiber vs global κ : depth upgrade from sewing

PROPOSITION 16.4 (κ *depth upgrade from sewing*). The fiber κ and global κ for $K3 \times E$ differ:

Level	κ	Shadow class
Single $K3$ fiber	24	$\mathbf{G}$ (Gaussian, $r_{\max} = 2$)
Global $K3 \times E$	5	$\mathbf{M}$ (mixed, $r_{\max} = \infty$)

The depth upgrade $\mathbf{G} \rightarrow \mathbf{M}$ occurs because sewing along E introduces the imaginary roots of $\mathfrak{g}_{\Delta_5}$ (multi-fiber bound states). The “lost” 19 units ($24 - 5 = 19$) enter the root structure: they represent the excess Hodge data absorbed by the Borchers product.

Four-path verification: (a) Borchers lift weight, (b) DMVV exponent, (c) bar Euler product matching, (d) shadow connection monodromy.

Verification: 78 tests in `k3e_relative_shadow.py`.

16.4 The Maulik–Okounkov R -matrix

PROPOSITION 16.5 (*MO R -matrix for $K3 \times E$*). The Maulik–Okounkov stable-envelope R -matrix for $K3 \times E$ matches the affine Yangian structure function $g(u)$ exactly. The Yang–Baxter equation is verified.

- (i) At the self-dual $K3$ limit (hyper-Kähler), the R -matrix is trivial: $g(u) = 1$.
- (ii) Nontrivial braiding requires breaking the hyper-Kähler structure via the Ω -background.

This is consistent with the E_2 mechanism: the self-dual $K3$ has E_∞ braiding (symmetric), and the Ω -deformation breaks it to E_2 .

Verification: 72 tests in `mo_rmatrix_k3e.py`.

17 $\chi_{\text{top}}/24 \neq \kappa$: the modular characteristic is categorical

The following result resolves a fundamental question about the $d = 3$ modular characteristic.

THEOREM 17.1 ($\chi_{\text{top}}/24 \neq \kappa$ in general). The topological Euler characteristic $\chi_{\text{top}}(X)/24$ does *not* equal the modular characteristic $\kappa(A_X)$ in general. Explicit counterexamples:

CY variety	$\chi_{\text{top}}/24$	κ	Match?
Elliptic curve E (CY ₁)	0	1	No
K ₃ surface (CY ₂)	1	2	No
$K3 \times E$ (CY ₃)	0	5	No
Resolved conifold	1/12	1	No
Quintic ($\chi = -200$)	-25/3	-25/3	Yes
$\mathbb{P}^5[3, 3]$ ($\chi = -144$)	-6	-6	Yes
$\mathbb{P}^5[2, 4]$ ($\chi = -176$)	-22/3	-22/3	Yes

The BCOV formula $\kappa = \chi_{\text{top}}/24$ holds for rigid compact CICYs (complete intersections in products of projective spaces) but fails systematically for K₃-fibered CY₃ and for non-compact toric CY₃.

Verification: 119 tests in `cy3_grand_atlas.py`.

PROPOSITION 17.2 (*The categorical Euler characteristic resolves the discrepancy*). The invariant controlling κ is the *categorical* Euler characteristic χ^{CY} , not the topological one. For $K3 \times E$: $\chi_{\text{top}} = 0$ but $\chi^{\text{CY}} = 5$.

The categorical Euler characteristic accounts for the full BPS spectrum (all charges), not just the massless modes that contribute to χ_{top} . For K₃-fibered CY₃, the infinite tower of bound states across fibers contributes to χ^{CY} via the Borchers denominator weight.

Five independent verifications of $\kappa(K3 \times E) = 5$:

- (a) Weight of the Igusa cusp form: $\text{weight}(\Delta_5) = 5$.
- (b) DMVV exponent: the Borchers lift weight formula gives $c(0)/2 = 10/2 = 5$.
- (c) Bar Euler product: the genus-1 coefficient of $\log Z_{\text{DT}}$ gives $F_1 = 5/24$.
- (d) Relative DT: fiber-by-fiber computation via K₃ fibers sewed along E .
- (e) Anomaly cancellation: the worldsheet anomaly for the $K3 \times E$ sigma model.

Verification: 78 tests in `k3e_relative_shadow.py`; 119 tests in `cy3_grand_atlas.py`.

Warning 17.3 (κ is not multiplicative). The modular characteristic is *not* multiplicative under products:

$$\kappa(K3) \cdot \kappa(E) = 2 \cdot 1 = 2, \quad \text{but} \quad \kappa(K3 \times E) = 5.$$

This is not a contradiction: κ measures the worldsheet anomaly of the full BPS spectrum, which includes bound states across fibers. The bound-state contributions (the “lost” 19 units of Proposition 16.4) account for the nonmultiplicativity.

18 Modularity constraints on κ

PROPOSITION 18.1 (*Integrality and positivity for BKM*). For a CY₃ to admit a genuine Siegel modular form as denominator identity (i.e., a genuine BKM superalgebra), the modular characteristic must satisfy $\kappa \in \mathbb{Z}_{>0}$ (positive integer weight for a holomorphic Siegel modular form). This rules out several families from naive BKM interpretation:

- Quintic: $\kappa = -25/3$ (negative, non-integer).
- Banana manifold (Schoen CY₃): $\kappa = 0$ ($\chi = 0$).

Only 9 of the 18 families in the atlas of Section 21 have integer κ .

Seven structural constraints for $K3 \times E$ are verified: integrality, positivity, Igusa weight match, Borcherds product convergence, row-sum vanishing of $\phi_{0,1}$, modular transformation law, and Hecke eigenform property. *Verification*: III tests in `cy3_modularity_constraints.py`.

Remark 18.2 ($\phi_{0,1}$ root/arity map). The $\phi_{0,1}$ Fourier coefficients classify BKM roots by arity in the shadow obstruction tower:

Root type	Discriminant	Shadow arity
Real ($c(-1) = 1$)	$D = -1$	Arity 2
Lightlike ($c(0) = 10$)	$D = 0$	Arity 3
Timelike ($c(D)$, signed)	$D > 0$	Arity ≥ 4

Negative values of $c(D)$ correspond to fermionic roots (odd generators of the BKM superalgebra). The total root multiplicity per BKM level for $K3 \times E$ is $24 = \chi(K3)$.

19 The toric CY₃ landscape

19.1 The conifold: wall-crossing as MC gauge equivalence

The resolved conifold $X = \mathcal{O}(-1) \oplus \mathcal{O}(-1) \rightarrow \mathbb{P}^1$ has charge lattice $\Gamma = \mathbb{Z}\gamma_1 \oplus \mathbb{Z}\gamma_2$ with $\langle \gamma_1, \gamma_2 \rangle = 1$. The two DT chambers (large-volume and flopped) are connected by the Kontsevich–Soibelman wall-crossing formula.

THEOREM 19.1 (*Conifold bar complex and wall-crossing*). (i) $\kappa(\text{conifold}) = 1$. Shadow class **G** (Gaussian, $r_{\max} = 2$).

(ii) Wall-crossing is a gauge transformation of the MC element:

$$\exp(\text{ad}_\alpha)(\Theta_I) = \Theta_{II},$$

where α lies in the lattice Lie algebra.

(iii) The pentagon identity is confirmed: $E(X)E(Y) = E(Y)E(XY)E(X)$, where $E(\cdot)$ denotes the quantum dilogarithm automorphism.

- (iv) The bar complexes in the two chambers are quasi-isomorphic:

$$H^*(B(\text{CoHA}_I)) \simeq H^*(B(\text{CoHA}_{II}))$$

as graded vector spaces, despite having different generators ($\dim B^1 = 2$ in chamber I vs $\dim B^1 = 3$ in chamber II). The extra generator in chamber II is exact (a bound state = bar boundary of a two-particle configuration).

Verification: 124 tests in `conifold_bar_complex.py`; 73 tests in `wallcrossing_gauge_engine.py`.

19.2 Local $\mathbb{P}^2$: the $\mathbb{Z}_3$ McKay quiver

PROPOSITION 19.2 (*Shadow tower of local $\mathbb{P}^2$*). For the total space $\mathcal{O}(-3) \rightarrow \mathbb{P}^2$:

- (i) $\kappa(\text{local } \mathbb{P}^2) = 3/2 = \chi(\mathbb{P}^2)/2$.
- (ii) Shadow class $\mathbf{M}$ (infinite depth).
- (iii) Gopakumar–Vafa invariant growth: $|n_d^0| \sim 27^d$ as $d \rightarrow \infty$, giving convergence radius $R = 1/27$ for the genus-0 prepotential.
- (iv) The E_1 -chiral algebra arises from the critical CoHA of the $\mathbb{Z}_3$ McKay quiver. The quiver has 3 vertices ($\mathbb{Z}_3$ representations), 9 arrows, and a potential W from the CY₃ condition.
- (v) Perturbative 5d CS reproduces the CoHA OPE: the deformed $\mathcal{W}_{1+\infty}$ acquires a loop correction factor $c_{\text{loop}} = -1/15$.

Verification: 77 tests in `local_p2_shadow.py`; 104 tests in `toric_shadow_engine.py`.

19.3 Local $\mathbb{P}^1 \times \mathbb{P}^1$: two-parameter shadow metric

PROPOSITION 19.3 (*Shadow tower of local $\mathbb{P}^1 \times \mathbb{P}^1$*). For $\mathcal{O}(-2, -2) \rightarrow \mathbb{P}^1 \times \mathbb{P}^1$:

- (i) $\kappa = 2 = h^{1,1}(\mathbb{P}^1 \times \mathbb{P}^1)$.
- (ii) The two-parameter shadow metric is $Q = \text{diag}(1, 1)$.
- (iii) The symmetric direction has shadow class $\mathbf{M}$ (infinite tower from the infinite BPS spectrum); the anti-symmetric direction has class $\mathbf{G}$.

Verification: 104 tests in `toric_shadow_engine.py`.

19.4 Perturbative comparison: 5d Chern–Simons

PROPOSITION 19.4 (*Perturbative 5d CS reproduces CoHA OPE*). The perturbative 5d noncommutative Chern–Simons theory (Costello–Li) reproduces the CoHA OPE structure for $\mathbb{C}^3$ and the conifold. Three-path verification:

$$\text{Feynman diagrams} = \text{shuffle algebra} = \mathcal{W}_{1+\infty} \text{ OPE.}$$

For the conifold, the perturbative answer is identical to $\mathbb{C}^3$; the difference is nonperturbative (wall-crossing).

Verification: 87 tests in `c3_envelope_comparison.py`; 94 tests in `toric_cy3_dt_engine.py`.

20 The E_2 bar complex

For an E_2 -chiral algebra, three distinct bar constructions are available, reflecting the three levels of operadic symmetry.

THEOREM 20.1 (*Three bar complexes and their comparison*).

Bar complex	Structure	Origin	Carries
$B_{E_1}^n(A)$	Ordered	Hochschild	Tensor product
$B_{E_2}^n(A)$	Braided	R -matrix	Braided tensor product
$B_{E_\infty}^n(A)$	Symmetric	Vol I shadow tower	S_n -quotient

The E_∞ bar complex is the S_n -quotient of the E_1 bar complex:

$$B_{E_\infty}^n(A) = (B_{E_1}^n(A))^{S_n}.$$

For $A = H_1$ (Heisenberg at level 1):

$$\dim B_{E_2}^n(H_1) = d(n) \quad (\text{divisor function}).$$

The E_2 bar complex is strictly between the E_1 and E_∞ bar complexes in size, and its Euler characteristics encode the E_2 -braided BPS spectrum.

Verification: 93 tests in `e2_bar_complex.py`; 110 tests in `e2_bar_cobar_koszul.py`.

Remark 20.2 (Relation to the Vol I shadow tower). The Vol I shadow tower uses the E_∞ bar complex (symmetric, no R -matrix data). The E_2 bar complex is the correct refinement for quantum vertex chiral groups: it retains the braiding information lost in the S_n -quotient. The passage $B_{E_2} \rightarrow B_{E_\infty}$ (forgetting the braiding) corresponds to the passage from the full quantum group $G(X)$ to its shadow data $\kappa, C, Q, \dots$

21 The grand atlas of CY_3 families

The following table collects the shadow tower data for all CY_3 families for which computations are available. The “Source” column records the formula or method used to determine κ .

CY_3 family	χ	$h^{1,1}$	$h^{2,1}$	κ	Class	Source
$\mathbb{C}^3$	—	—	—	1	G	$\kappa(H_1) = 1$
Conifold	—	1	0	1	G	DT genus-1
Local $\mathbb{P}^2$	—	1	0	3/2	M	GV free energy
Local $\mathbb{P}^1 \times \mathbb{P}^1$	—	2	0	2	M/G	$h^{1,1}$
$K3 \times E$	0	21	21	5	M	$\text{wt}(\Delta_5)$
Enriques $\times E$	0	11	11	4	M	Allcock form
Quintic	−200	1	101	−25/3	M	$\chi/24$
$\mathbb{P}^5[3, 3]$	−144	1	73	−6	M	$\chi/24$
$\mathbb{P}^5[2, 4]$	−176	1	89	−22/3	M	$\chi/24$
$\mathbb{P}^5[2, 2, 3]$	−144	1	73	−6	M	$\chi/24$

CY ₃ family	χ	$h^{1,1}$	$h^{2,1}$	κ	Class	Source
$\mathbb{P}^7[2, 2, 2, 2]$	-128	1	65	-16/3	M	$\chi/24$
Banana	0	2	0	0	$\geq G$	BKY
K ₃ -fibered (general)	var	var	var	var	M	relative DT
Borcea–Voisin	var	var	var	$\chi/24$	M	involution

Observation 21.1 (Atlas patterns). (i) The BCOV formula $\kappa = \chi_{\text{top}}/24$ holds for all rigid CICYs but fails for K₃-fibered families (Theorem 17.1).

(ii) For toric CY₃ with N vertices in the toric web diagram, the shadow depth satisfies $r_{\text{max}} \leq N$ (conjectural toric vertex bound).

(iii) All compact CY₃ with $\chi \neq 0$ are class **M**.

(iv) Only 9 of the 14+ catalogued families have integer κ . A genuine BKM algebra requires $\kappa \in \mathbb{Z}_{>0}$ (Proposition 18.1).

(v) The Enriques $\times E$ modular characteristic $\kappa = 4$ is *not* half the K₃ value; the ratio $\kappa(K3 \times E)/\kappa(\text{Enr} \times E) = 5/4$ reflects the change in automorphic weight from Δ_5 (weight 5, $\text{Sp}_4(\mathbb{Z})$) to Allcock’s form (weight 4, $O(2, 10)$).

Verification: 119 tests in `cy3_grand_atlas.py`; 72 tests in `enriques_shadow.py`.

22 Physical applications of the shadow–BPS correspondence

22.1 BPS black hole entropy from the shadow tower

PROPOSITION 22.1 (*Shadow tower entropy expansion*). The BPS black hole entropy admits an expansion organized by shadow arity:

- (i) *Arity 2 = Bekenstein–Hawking:* $S_{\text{BH}} = \pi\sqrt{2\kappa \cdot D}$, where D is the discriminant of the charge vector. This is the leading-order entropy, controlled by κ alone.
- (ii) *Arity 3 = logarithmic correction:* $\delta S_3 = -(27/4) \log D + O(1)$.
- (iii) *Arity ≥ 4 = power corrections:* controlled by the quartic shadow and higher, giving Bessel-type corrections $\sim D^{-k/2}$ for $k \geq 1$.

The shadow class controls the qualitative entropy structure:

Shadow class	Black hole entropy
G	Bekenstein–Hawking only (no quantum corrections beyond genus 1)
L	Bekenstein–Hawking + logarithmic correction
M	Genuine black holes with infinite tower of quantum corrections

The passage from Bekenstein–Hawking to the full quantum-corrected entropy is the passage from $\Theta^{\leq 2}$ to Θ_A in the shadow obstruction tower.

22.2 Calogero–Moser/Bethe ansatz dictionary

PROPOSITION 22.2 (*Integrable system/shadow tower dictionary*). For the $\mathcal{W}_N$ chiral algebra, the Calogero–Moser integrals of motion I_k correspond to shadow tower components:

Integrable system	Shadow tower
I_2 (quadratic Hamiltonian)	κ (modular characteristic)
I_3 (cubic integral)	C (cubic shadow)
I_4 (quartic integral)	Q (quartic resonance class)

The Bethe ansatz equations for the quantum Calogero–Moser system are the saddle-point equations of the shadow generating function.

22.3 Lattice model finite-size corrections

PROPOSITION 22.3 (*Finite-size corrections from the shadow tower*). Lattice model finite-size corrections on a system of size L decay at a rate exactly $q = e^{-L/\xi}$, where ξ is the correlation length. The boundary effects encode shadow tower data: the leading correction is proportional to κ/L^2 , subleading to C/L^3 . The MacMahon box formula (counting plane partitions in an $a \times b \times c$ box) is controlled by the shadow tower of $\mathcal{W}_{1+\infty}$.

23 Five routes to the chiral algebra

23.1 The Costello–Li chain for $\mathbb{C}^3$

The Costello–Li programme provides the most explicit route from CY₃ geometry to chiral algebra. For $\mathbb{C}^3$, the chain is:

$$5d \text{ NCCS on } \mathbb{C}^3 \rightarrow Y^+(\widehat{\mathfrak{gl}}_1) \simeq \mathcal{W}_{1+\infty}.$$

Five independent routes to $\mathcal{W}_{1+\infty}$:

- (a) *Critical CoHA*: Kontsevich–Soibelman $\rightarrow$ Schiffmann–Vasserot $\rightarrow Y^+(\widehat{\mathfrak{gl}}_1)$.
- (b) *Crystal melting*: counting plane partitions $\rightarrow$ MacMahon $\rightarrow$ vertex operators.
- (c) *Shuffle algebra*: explicit generators and relations from the CoHA shuffle product.
- (d) *Factorization envelope*: $PV^*(\mathbb{C}^3) \rightarrow$ Lie conformal algebra $\rightarrow$ Nishinaka envelope.
- (e) *5d Chern–Simons*: Costello–Li perturbative computation reproduces the OPE.

At $N = 1$ (single D-brane), all five produce the Heisenberg VOA H_1 .

23.2 Five routes for $K3 \times E$

Five routes to the chiral algebra of $K3 \times E$ have been mapped:

- (a) *Relative DT*: fiber K_3 shadow data sewed along E via DMVV.
- (b) *Borcherds BKM*: root multiplicities from $\phi_{0,1} \rightarrow U(\mathfrak{n}_+(\mathfrak{g}_{\Delta_5}))$.
- (c) *Maulik–Okounkov*: stable envelopes on $\text{Hilb}^n(K3 \times E) \rightarrow$ Yangian R -matrix.

- (d) *Vafa–Witten*: partition function on $K3 \rightarrow$ mock modular forms $\rightarrow$ chiral characters.
- (e) *Relative chiral algebra*: direct construction via $K3$ fibration (Proposition 16.2).

23.3 The E_1 mechanism: Dunn additivity and Ω -background

The reduction from E_2 to E_1 in the presence of the Ω -background has a clean operadic explanation. By Dunn additivity:

$$E_2 \simeq E_1 \otimes_{E_0} E_1.$$

The Ω -background freezes one of the two E_1 -factors (it introduces a preferred direction in the plane), reducing the native E_2 to E_1 . The Drinfeld center passage (Theorem 14.5) restores the E_2 structure by recovering the frozen factor.

This explains why the CoHA is only E_1 : it is the Ω -deformed theory, with one E_1 -factor frozen. The full E_2 -chiral algebra A_X is the Drinfeld center, which unfreezes it.

24 Cross-volume bridges

The results of Sections 14–23 connect the CY_3 programme (Vol III) to the algebraic engine (Vol I) and the holomorphic-topological QFT framework (Vol II).

24.1 Shadow tower identification

The shadow obstruction tower Θ_A of Vol I (Theorems A–D, the bar-intrinsic construction $\Theta_A := D_A - d_0$) specializes as follows for CY_3 chiral algebras:

Vol I object	CY_3 specialization	Identification
$\kappa(A)$	$\kappa(\mathbb{C}^3) = 1$	MacMahon genus-1
Cubic shadow C	BKM lightlike roots	$c(0) = 10$ for $K3 \times E$
Quartic shadow Q	BKM timelike roots	$c(D > 0)$ for $K3 \times E$
Shadow class $G/L/C/M$	Toric vertex bound	$r_{\max} \leq N$
Bar Euler product	Denominator identity	$1/M(q)$ for $\mathbb{C}^3$; Δ_5 for $K3 \times E$
Shadow connection ∇^{sh}	Picard–Fuchs	Modular family
Complementarity $Q(A) + Q(A^\dagger)$	Koszul dual CY_3	Mirror symmetry

24.2 Swiss-cheese and the E_2 bar complex

The Vol II Swiss-cheese structure $SC^{\text{ch}, \text{top}}$ (Theorem H) provides the operadic framework for the E_2 bar complex of Section 20. The three bar complexes $(B_{E_1}, B_{E_2}, B_{E_\infty})$ correspond to the three sectors:

- B_{E_1} : the open sector (Hochschild, ordered tensors).
- B_{E_2} : the boundary sector (braided, with R -matrix data).
- B_{E_∞} : the closed sector (symmetric, Vol I shadow tower).

The $E_1 \rightarrow E_2$ passage via Drinfeld center (Theorem 14.5) corresponds to the open-to-bulk passage in Vol II via the chiral derived center $\mathcal{Z}_{\text{ch}}^{\text{der}}(A)$.

24.3 The completed modular Koszul datum for CY₃

The eight-fold completed modular Koszul datum $\Pi_X^{\text{oc}}(A)$ of Vol I specializes to the CY₃ setting as:

$$\Pi_X^{\text{oc}}(A_{X_3}) = (\mathcal{F}_X(A), \overline{B}_X(A), \Theta_A, L_A, (V^{\text{br}}, T^{\text{br}}), R_4^{\text{mod}}, \mathcal{Z}_{\text{ch}}^{\text{der}}(A), \text{Tr}_A),$$

where the BKM root system supplies the root lattice structure and the Borchers denominator formula controls the bar Euler product.

Verification: 124 tests in `cross_volume_shadow_bridge.py`.

25 Updated status of the programme

The results of Sections 14–24 substantially advance the programme relative to the baseline recorded in Section 1.

Target	Status	Evidence
CY-A ($d = 3$ functor)	Verified, toric	Functor chain for $\mathbb{C}^3$ verified end-to-end (§14). $\mathbb{S}^3$ -framing trivial (§15).
CY-B (E_2 -chiral KD)	Verified, $\mathbb{C}^3$, $K3 \times E$	Bar Euler product = $1/M(q)$ (§14.3). BKM denominator (§16.1).
CY-C (Quantum group)	Verified, $\mathbb{C}^3$	Drinfeld center gives E_2 -braided category with Yang R -matrix (§14.2).
CY-D (Modular char.)	Established	κ computed for 14+ families (§21).
$\mathbb{S}^3$ -framing	Topol. trivial	$\pi_3(B\text{Sp}) = \pi_3(BU) = 0$ (§15).
$E_1 \rightarrow E_2$	Trivial, all tested	Cor. 15.2.
$\chi/24 \neq \kappa$	Resolved	Thm. 17.1: categorical, not topological.

25.1 Remaining gaps

1. **CY-A for compact CY₃:** the $\mathbb{S}^3$ -framing is topologically trivializable, but the chain-level construction of the trivialization remains open.
2. **κ formula for $K3$ -fibered CY₃:** the formula $\kappa = \chi^{\text{CY}}$ is verified for $K3 \times E$ and Enriques $\times E$ but is conjectural beyond these.
3. **Non-BKM families:** CY₃ with non-integer κ do not admit naive BKM interpretations.
4. **Full motivic DT:** the shadow tower = BKM roots identification is at the level of generating series; the motivic upgrade (AP42) remains open.

26 Updated open questions

The original open questions (Section 7) are updated in light of the new results:

1. **Root datum of the quintic:** still open. $\kappa = -25/3$ is non-integer and negative (Proposition 18.1).
2. **$\mathbb{S}^3$ -framing construction:** *partially resolved* — topological obstruction vanishes universally (Theorem 15.1); chain-level via holomorphic CS is the remaining step.

3. **Automorphic form for Hitchin/ E_8** : still open.
4. **Wall-crossing as MC gauge equivalence**: *verified for the conifold* (Theorem 19.1).
5. **Is $\kappa = h^{1,1}/4$ universal?**: *No* — the correct invariant is $\kappa = \text{wt}(\text{denominator form})$, not a function of Hodge numbers alone.
6. **Shadow depth for quantum vertex chiral groups**: *characterized* (Observation 21.1).
7. E_2 **bar complex**: *constructed and computed* (Theorem 20.1).
8. $\chi_{\text{top}}/24 \neq \kappa$: *resolved* (Theorem 17.1).
9. **Quantum geometric Langlands as E_2 -chiral Koszul duality**: still open.
10. **Eight diagonal divisor Siegel modular forms**: still open.

27 Compute verification summary

The results recorded above are backed by 5,010 tests across 60+ compute modules (all passing). The following table lists the modules most directly relevant to the frontier results.

Module	Tests	Primary result
c3_grand_verification	115	$\mathbb{C}^3$ Rosetta Stone, $\kappa = 1$ five-path
conifold_bar_complex	124	Conifold bar complex, wall-crossing
cy_to_chiral_functor	124	$d = 3$ functor chain, $\mathbb{S}^3$ -framing
cross_volume_shadow_bridge	124	Cross-volume verification
cy3_grand_atlas	119	Grand atlas, 14+ CY_3 families
cy3_modularity_constraints	111	Modularity constraints on κ
e2_bar_cobar_koszul	110	E_2 bar-cobar-Koszul
toric_shadow_engine	104	Toric shadow tower landscape
c3_shadow_tower	98	$\mathbb{C}^3$ shadow tower, class G
cy_bar_complex_engine	97	General CY bar complex
c3_lie_conformal	95	$\text{GL}(3)$ -invariant brackets vanish
toric_cy3_dt_engine	94	Toric DT engine
drinfeld_center_yangian	93	Drinfeld center, R -matrix, YBE
e2_bar_complex	93	E_2 bar complex, $\dim = d(n)$
coha_drinfeld_bulk	93	CoHA, Drinfeld center, bulk
k3e_coha_structure	90	$K3 \times E$ CoHA, BKM roots
bkm_shadow_complete	88	BKM shadow tower complete
c3_envelope_comparison	87	Envelope/shuffle/crystal comparison
k3e_relative_shadow	78	Relative DT, fiber vs global κ
local_p2_shadow	77	Local $\mathbb{P}^2$, $\kappa = 3/2$
wallcrossing_gauge_engine	73	Wall-crossing gauge equivalence
mo_rmatrix_k3e	72	MO R -matrix for $K3 \times E$

Module	Tests	Primary result
enriques_shadow	72	Enriques $\times E$, $\kappa = 4$
cy3_hochschild	71	HH/HC computation, BTT
banana_shadow	63	Banana manifold, $\kappa = 0$
bar_comparison_c3	59	Bar Euler product = $1/M(q)$
macmahon_shadow_decomposition	50	MacMahon shadow non-decomposition